GEN AI

GEN AI & Agentic AI

Detailed Topic-Wise Curriculum & Hour Allocation

Major topic → sub-topic → hours → detailed classroom/LMS coverage

80	80	70	146
Month 1 — Live	Month 2 — Live	Month 3 — Live	Self-Paced

Grand Total **376** hours

How to Read This Document

For example: Python = 20 hours, and the table underneath distributes those 20 hours across the actual topics that should be taught.

01

Foundations

Month 1 — Data & ML Foundations

80 hours — Live Classroom

Detailed sub-topic allocation with exact module totals.

Python

20 hours

Python environment & programming basics Jupyter/IDE workflow, variables, data types, input/output, operators and basic debugging.	2 hrs
Control flow if/elif/else, nested conditions, for/while loops, range, break and continue.	2 hrs
Functions Defining functions, parameters, return values, scope, default/keyword arguments and reusable code.	3 hrs
Core data structures Lists, tuples, sets and dictionaries; indexing, slicing, methods and practical data manipulation.	4 hrs
Strings & text handling String operations, formatting, common methods, parsing and basic text processing.	2 hrs
Modules, packages & files Imports, modules, packages, paths and reading/writing text and CSV files.	2 hrs
Exceptions & debugging try/except/finally, common errors, debugging strategy and defensive programming.	2 hrs
OOP foundations Classes, objects, constructors, attributes, methods and basic inheritance.	2 hrs
Practical Python for data work Small data-processing exercise combining functions, collections and files.	1 hr
Module total	**20 hours**

SQL

12 hours

Relational database foundations Tables, rows, columns, primary keys, relationships and basic database terminology.	1 hr
SELECT & filtering SELECT, aliases, WHERE, comparison/logical operators, IN, BETWEEN and LIKE.	2 hrs
Sorting, limiting & aggregation ORDER BY, LIMIT, COUNT, SUM, AVG, MIN, MAX and GROUP BY.	2 hrs
HAVING & conditional analysis HAVING, CASE expressions and grouped business questions.	1 hr
Joins INNER, LEFT, RIGHT and practical multi-table joins; avoiding duplicate rows.	3 hrs
Subqueries & CTEs Nested queries, correlated-query intuition and common table expressions.	1.5 hrs
Window functions PARTITION BY, ORDER BY, ROW_NUMBER, RANK, DENSE_RANK and analytical use cases.	1.5 hrs
Data cleaning queries NULL handling, COALESCE, duplicate detection and basic transformation queries.	1 hr
Module total	**12 hours**

Statistics

16 hours

Descriptive statistics Population vs sample, mean, median, mode, range, variance and standard deviation.	3 hrs
Percentiles & distributions Quartiles, IQR, percentiles, frequency distributions and interpreting spread.	2 hrs
Probability foundations Sample spaces, events, conditional probability, independence and multiplication/addition rules.	3 hrs
Bayes theorem Prior, likelihood, evidence and posterior; intuitive applications to classification.	2 hrs
Random variables & expectation Discrete/continuous variables, probability distributions, expected value and variance.	2 hrs
Common distributions Bernoulli, Binomial, Normal and related distribution intuition.	2 hrs
Correlation & covariance Measuring relationships, covariance vs correlation and interpretation limits.	1 hr
Sampling & basic inference Sampling concepts, sampling error, confidence-interval intuition and hypothesis-testing overview.	1 hr
Module total	**16 hours**

ML Foundations

24 hours

ML overview & workflow Supervised/unsupervised learning, features/targets, training workflow and common use cases.	2 hrs
Data preprocessing Missing values, categorical encoding, scaling, outliers and train/validation/test splits.	3 hrs
Linear regression Prediction, loss, coefficients, assumptions, evaluation and practical implementation.	3 hrs
Logistic regression Classification, sigmoid, decision boundary, log loss and probability-based prediction.	3 hrs
k-Nearest Neighbors Distance, neighborhood selection, classification/regression and scaling considerations.	2 hrs
Decision trees Splits, impurity, depth, overfitting and interpretability.	3 hrs
Ensemble basics Bagging, random forests, boosting intuition and why ensembles improve performance.	2 hrs
Model evaluation Regression and classification metrics, confusion matrix, precision, recall, F1 and ROC-AUC.	3 hrs
Cross-validation & tuning Cross-validation, hyperparameters, grid/random search and avoiding data leakage.	2 hrs
Mini ML project End-to-end dataset workflow from preprocessing through evaluation and interpretation.	1 hr
Module total	**24 hours**

Review / Practice Buffer

8 hours

Python practice Coding exercises and debugging.	2 hrs
SQL practice Query-building and analytical problem solving.	2 hrs
Statistics practice Numerical/statistical exercises and interpretation.	2 hrs
ML revision Model selection, evaluation and mini case study.	2 hrs
Module total	**8 hours**

02

Deep Learning & GenAI

Month 2 — Advanced ML, Deep Learning & GenAI/LLMs

Detailed sub-topic allocation with exact module totals.

80 hours — Live Classroom

Advanced ML — 10 hours

Feature engineering Creating useful features, transformations, encoding strategies and leakage awareness.	2 hrs
Regularization L1/L2 regularization, bias-variance trade-off and controlling overfitting.	1.5 hrs
SVM Margins, kernels, support vectors and classification intuition.	1.5 hrs
Clustering k-Means, cluster interpretation, distance and choosing cluster count.	2 hrs
Dimensionality reduction PCA intuition, variance preservation and visualization.	1.5 hrs
Advanced model selection Comparing models, tuning strategy and practical evaluation.	1.5 hrs
Module total	10 hours

Deep Learning — 16 hours

Neural-network foundations Neurons, layers, weights, biases and forward propagation.	2 hrs
Activation functions Sigmoid, tanh, ReLU and why nonlinear activation is required.	2 hrs
Loss functions Regression/classification losses and how loss guides learning.	2 hrs
Gradient descent Gradients, learning rate, optimization intuition and parameter updates.	2 hrs
Backpropagation intuition Chain rule intuition and how error signals flow backward.	2 hrs
Deep neural networks Hidden layers, capacity, overfitting and practical training workflow.	2 hrs
Improving neural networks Initialization, normalization, dropout and regularization concepts.	2 hrs
Deep-learning mini project Build, train, evaluate and interpret a small neural network.	2 hrs
Module total	16 hours

PyTorch — 14 hours

PyTorch tensors Tensor creation, shapes, indexing, operations and device concepts.	2 hrs
Datasets & DataLoaders Dataset abstraction, batching, shuffling and input pipelines.	2 hrs
Autograd Computational graphs, gradients and automatic differentiation.	2 hrs
Building models with nn.Module Layers, forward method, model structure and reusable architecture.	3 hrs
Training loop Optimizer, loss, forward pass, backward pass, update and epoch workflow.	3 hrs
Evaluation & saving models Evaluation mode, metrics, checkpoints and loading trained models.	1 hr
PyTorch mini project Train a practical model and document the workflow.	1 hr
Module total	14 hours

GenAI + LLMs — 12 hours

Generative AI foundations Generative vs predictive AI, modalities, model families and application landscape.	2 hrs
LLM architecture overview Tokens, embeddings, transformer intuition, attention and next-token prediction.	3 hrs
Training & inference Pretraining, instruction tuning, inference, context windows and temperature.	2 hrs
LLM capabilities & limitations Hallucination, context limitations, knowledge boundaries and reliability concerns.	2 hrs
Fine-tuning concepts Fine-tuning, parameter-efficient approaches and when adaptation is useful.	1.5 hrs
LLM application patterns Chat, extraction, summarization, classification and generation workflows.	1.5 hrs
Module total	12 hours

Prompt Engineering — 8 hours

Prompt fundamentals Instructions, context, constraints and output requirements.	2 hrs
Few-shot & structured prompting Examples, schemas, formatting and reliable output patterns.	2 hrs
Reasoning-oriented prompting Breaking complex tasks into steps without exposing private chain-of-thought.	1 hr
Prompt evaluation & iteration Test cases, failure analysis and prompt refinement.	1 hr
Practical prompt project Build a structured LLM workflow and evaluate outputs.	2 hrs
Module total	8 hours

Embeddings + Vector DB — 10 hours

Vector representations Embeddings, semantic meaning and vector-space intuition.	2 hrs
Similarity measures Cosine similarity, Euclidean distance and semantic nearest-neighbor retrieval.	2 hrs
Vector databases Collections/indexes, vectors, metadata and retrieval concepts.	2 hrs
Chunking & metadata Document chunking, metadata design and retrieval-oriented preprocessing.	2 hrs
Semantic search mini project Create embeddings, store/search vectors and inspect retrieved results.	2 hrs
Module total	10 hours

RAG — 10 hours

RAG architecture Ingestion, chunking, embedding, retrieval, prompting and generation.	2 hrs
Document processing Loading documents, cleaning, chunking and metadata.	2 hrs
Retrieval strategies Similarity search, top-k, filters, hybrid retrieval intuition and reranking overview.	2 hrs
Grounded generation Passing retrieved context to an LLM and reducing unsupported answers.	2 hrs
RAG mini project & evaluation Build a small document QA system and test retrieval/answer quality.	2 hrs
Module total	10 hours

03

Agents & Deployment

Month 3 — Agents, APIs, Deployment & Projects

70 hours — Live Classroom

Detailed sub-topic allocation with exact module totals.

Agents (LangChain/LangGraph + Tool Calling)

14 hours

Agent concepts & architecture Agents vs chains, state, goals, planning concepts and execution loops.	2 hrs
Tool/function calling Tool schemas, function arguments, tool selection and handling tool results.	2 hrs
LangChain fundamentals Models, prompts, tools, chains and practical application structure.	2 hrs
LangGraph Graph-based agent workflows, nodes, edges, state and controlled execution.	3 hrs
Memory & state Conversation state, short-term memory and managing context.	2 hrs
Agent reliability & safety Tool validation, retries, guardrails and failure handling.	1.5 hrs
Agent project Build an agent that uses multiple tools and document/context retrieval.	1.5 hrs
Module total	**14 hours**

Application Development (FastAPI/REST APIs)

14 hours

HTTP & REST fundamentals Requests/responses, methods, status codes, headers and REST design.	2 hrs
FastAPI basics Application structure, routes, path/query parameters and request handling.	3 hrs
Pydantic & validation Schemas, validation, serialization and API contracts.	2 hrs
Integrating ML/LLM models Calling trained models/LLMs from APIs and returning structured responses.	3 hrs
Authentication & error handling API authentication concepts, exceptions and consistent error responses.	2 hrs
Testing & API documentation Testing endpoints, OpenAPI/Swagger and practical API verification.	2 hrs
Module total	**14 hours**

Docker + Deployment

14 hours

Containers & Docker concepts Images, containers, registries and container lifecycle.	2 hrs
Dockerfiles Base images, dependencies, working directories, commands and reproducible builds.	3 hrs
Ports, volumes & networking Port mapping, persistent data and container communication.	2 hrs
Docker Compose Multi-container applications, services, environment variables and dependencies.	2 hrs
Containerizing an AI application Package FastAPI/model application into a deployable container.	3 hrs
Deployment workflow Build, tag, push and deploy; environment configuration and basic troubleshooting.	2 hrs
Module total	**14 hours**

Cloud

8 hours

Cloud fundamentals Compute, storage, networking and managed-service concepts.	1.5 hrs
Deployment environments Local vs staging vs production and configuration management.	1.5 hrs
Cloud storage Object storage concepts and application integration.	1 hr
Scaling & reliability basics Horizontal scaling, stateless services, health checks and availability concepts.	2 hrs
Cloud cost & security basics Resource sizing, secrets, least privilege and cost-awareness.	2 hrs
Module total	**8 hours**

End-to-End Projects (3–4, incl. one deployed)

14 hours

Project planning & architecture Problem definition, requirements, architecture, dataset/source selection and milestones.	2 hrs
ML project Data preparation, modeling, evaluation and business interpretation.	3 hrs
GenAI/RAG project Document ingestion, embeddings, retrieval, LLM generation and evaluation.	3 hrs
Agent project Tool calling, state, agent workflow and API integration.	3 hrs
Deployment & portfolio packaging Containerize/deploy one project, README, architecture diagram and demo preparation.	3 hrs
Module total	**14 hours**

Placement Preparation

6 hours

Technical interview preparation Python/SQL/ML/GenAI question practice and concept revision.	2 hrs
Project storytelling Explain problem, approach, architecture, metrics, trade-offs and contribution.	2 hrs
Coding & mock interview practice Timed coding, debugging, ML case questions and mock interview feedback.	2 hrs
Module total	**6 hours**

04

Self-Paced

Self-Paced Track — Mathematics, Theory & MLOps

146 hours — LMS / Self-Paced

Detailed sub-topic allocation with exact module totals.

Mathematics for ML

30 hours

Linear algebra fundamentals	Vectors, matrices, matrix operations, dot product, norms, projections and geometric intuition.	8 hrs
Calculus fundamentals	Functions, derivatives, partial derivatives, gradients, chain rule and optimization intuition.	8 hrs
Probability foundations	Probability rules, conditional probability, Bayes theorem and random variables.	5 hrs
Optimization basics	Loss functions, gradient-based optimization and local/global minima intuition.	5 hrs
Math for practical ML	Connecting algebra/calculus concepts to regression, logistic regression and neural networks.	4 hrs
Module total		30 hours

Probability & Statistics

28 hours

Descriptive statistics	Central tendency, dispersion, percentiles and distributions.	4 hrs
Probability	Events, conditional probability, independence and Bayes theorem.	5 hrs
Random variables & distributions	Discrete/continuous variables, expectation, variance and common distributions.	5 hrs
Sampling & estimation	Sampling methods, sampling distributions and estimation intuition.	4 hrs
Confidence intervals	Interval estimates and interpretation.	3 hrs
Hypothesis testing	Null/alternative hypotheses, p-values, Type I/II errors and test selection.	4 hrs
Correlation & regression statistics	Covariance, correlation and statistical interpretation of relationships.	3 hrs
Module total		28 hours

ML Theory / Math Behind Algorithms

26 hours

Linear regression mathematics	Least squares, loss minimization and coefficient interpretation.	4 hrs
Logistic regression mathematics	Sigmoid, log-odds, likelihood and log loss.	4 hrs
Distance-based algorithms	Euclidean/Manhattan distance, scaling and geometric intuition.	3 hrs
Decision-tree mathematics	Entropy, information gain and Gini impurity.	3 hrs
Regularization & bias-variance	Underfitting, overfitting, regularization and generalization.	4 hrs
Probability-based classification	Likelihood, priors, posterior reasoning and Naive Bayes intuition.	3 hrs
Model evaluation theory	Metric definitions, threshold effects and validation principles.	3 hrs
Optimization & generalization review	Gradient descent intuition and why validation matters.	2 hrs
Module total		26 hours

Deep Learning Math

24 hours

Vectors & matrix transformations	Tensor shapes, linear transformations and neural-network layer operations.	4 hrs
Derivatives & partial derivatives	Derivative rules and multivariable differentiation.	4 hrs
Gradients & chain rule	Gradient vectors, chain rule and backpropagation intuition.	5 hrs
Optimization	Gradient descent, learning rates, momentum and optimization landscape.	4 hrs
Loss functions	MSE, binary cross-entropy, categorical cross-entropy and gradient implications.	3 hrs
Regularization mathematics	L1/L2 penalties and parameter constraints.	2 hrs
Deep-learning math review	Connect the mathematics to a complete training step.	2 hrs
Module total		24 hours

GenAI / LLM Theory

22 hours

Tokens & embeddings	Tokenization, embedding vectors and semantic representation.	3 hrs
Transformers	Self-attention, query/key/value intuition, positional information and transformer blocks.	5 hrs
LLM training	Pretraining objective, next-token prediction, data and compute considerations.	4 hrs
Instruction tuning & alignment	Instruction tuning, preference optimization concepts and RLHF overview.	3 hrs
Inference	Decoding, temperature, top-k/top-p, context windows and latency concepts.	3 hrs
Evaluation & limitations	Hallucinations, benchmarks, factuality and application-level evaluation.	2 hrs
LLM system concepts	Model APIs, context management and high-level production considerations.	2 hrs
Module total		22 hours

MLOps (Theory + Hands-on)

16 hours

MLOps Overview & Lifecycle	ML lifecycle stages and why MLOps matters.	1.5 hrs
MLOps Architecture Concepts	Conceptual architecture for repeatable ML pipelines and environments.	2.5 hrs
CI/CD for ML Pipelines	Automated testing, build and deployment concepts for ML workflows.	3.5 hrs
Model Versioning (MLflow/DVC)	Experiment tracking, model artifacts, versions and reproducibility.	3.5 hrs
Monitoring & Logging Basics	Model performance, drift, logs and production observability.	3 hrs
Cloud Architecture Concepts	Conceptual cloud patterns for ML; practical cloud deployment is covered live.	2 hrs
Module total		16 hours

FF

End to end

Final Flow

The complete learning journey from foundation to deployment and placement.

01 **Python + SQL**
Programming and data-access foundation

02 **Statistics + Mathematics**
Quantitative foundation for ML

03 **Machine Learning**
Core predictive and unsupervised learning

04 **Deep Learning + PyTorch**
Neural networks and practical DL

05 **GenAI + LLMs**
Modern generative AI foundations

06 **Prompt Engineering**
Reliable and structured LLM interaction

07 **Embeddings + Vector DB**
Semantic representation and retrieval

08 **RAG**
Grounded LLM applications

09 **Agents**
Tool-using and stateful AI workflows

10 **FastAPI + REST**
Turn models into applications

11 **Docker + Cloud**
Package and deploy applications

12 **Projects + Placement**
Portfolio, interviews and career readiness